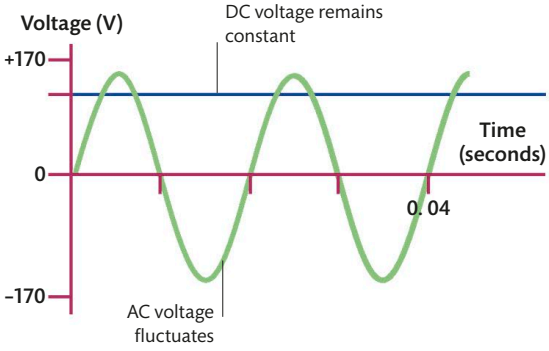
Generators

Electrical generators work using the principle of electromagnetic induction. As a wire coil spins between the two poles of a magnet, an electric current is induced to flow through the wire and around a circuit.

Direct current and alternating current

Generators produce either alternating or direct electric current. Direct current (DC) flows in one direction only around an electrical circuit and is generated by batteries and cells. Alternating current (AC) reverses its direction many times per second. Its voltage can also be greatly increased or decreased by devices called transformers, so it travels more efficiently over distances, which is why AC is used for our main power grid.

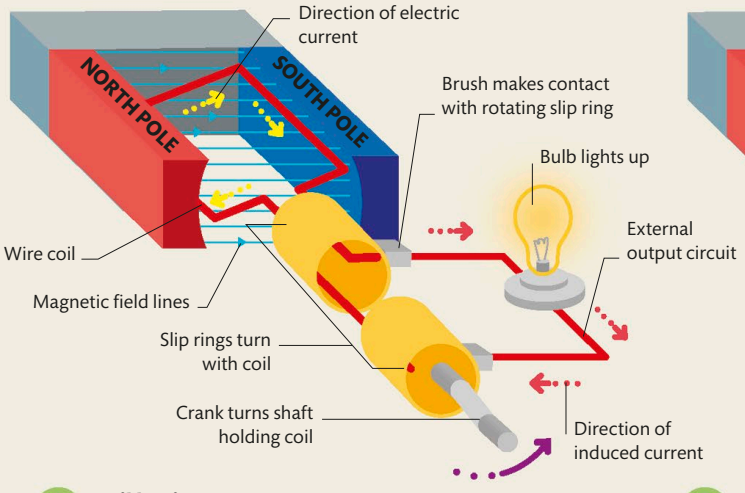


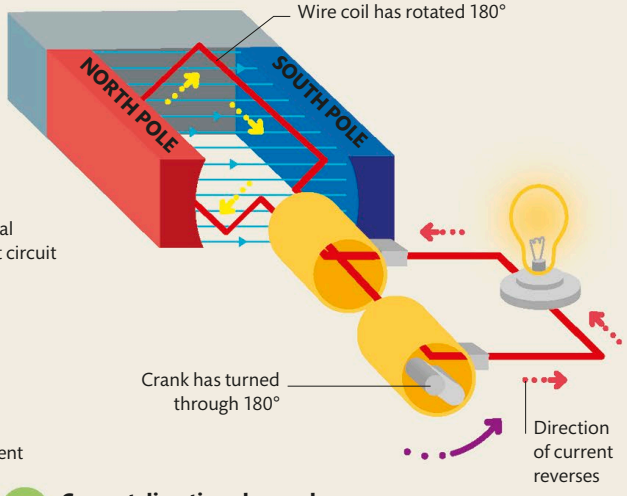
Contrasting current
DC produces a steady voltage, while AC's voltage alters as it reverses direction. AC's voltage must rise higher than DC's to transfer the same energy during the same time period.

KEY
AC
DC

AC generators

An AC generator is also known as an alternator. Its rotating wire coil is connected to an electric output circuit via slip rings and brushes. The brushes make continual contact with the output circuit, conducting current between the rotating slip ring and the fixed wires attached to the brushes. The current induced in an AC generator changes direction twice during each 360° revolution completed by the coil.





1 Coil begins to turn
The shaft of this experimental AC generator is rotated by mechanical force delivered by turning a handle. The shaft turns a wire coil through a magnetic field generated by a permanent magnet's north and south poles. As the coil cuts through the magnetic field, a current flowing in one direction is induced, reaching a peak when the coil lies horizontally through the field.

2 Current direction changed
As the coil completes another 180° turn through the magnetic field, the points that were initially face up are now face down; the coil's position relative to the north and south magnetic poles alters, its magnetic polarity changes, and the direction of the induced electric current reverses. The current reverses every half-turn, flowing to the slip rings and brushes and into the external output circuit.